ENTERPRISE AUTOMATION SYSTEM

GST Registration & Multi-State Automation Platform

An End-to-End Enterprise Solution for GST Client Onboarding, Intelligent Document OCR, Real-Time Human-in-the-Loop Bot Automation, and Multi-State Master TRN Synchronization.

🚀 Multi-State & Regular Automation

Full support for single-state GST registration as well as nationwide Multi-State Registration (/registration/msr).

🤖 Real-time Playwright Bot

Headless browser automation running on system Chrome/Edge with live status streaming and automated field injection.

🔐 Human-in-the-Loop CAPTCHA & OTP

Socket.io real-time streaming allows operators to solve portal CAPTCHAs and Aadhaar/TRN OTPs on the fly.

📄 Client OCR & Cloud Sync

Client-side Tesseract.js OCR for instant PAN and Aadhaar extraction, synced with cloud Supabase database.

Project Name: GST Registration Automation Platform
Architecture: Express.js + Vite/Vanilla JS + Playwright + Supabase

Documentation Version: 2.0 (Stable)
Date: October 2026

1. Executive Summary & Problem Statement

The Goods and Services Tax (GST) registration process in India is notoriously tedious and resource-intensive for chartered accountants, tax consultants, and enterprise compliance teams. A typical single application requires filling out over 50 data fields across 10 distinct tabs (Business Details, Promoters, Authorized Signatories, Principal Place of Business, HSN/SAC codes, etc.), accompanied by frequent portal timeouts, session expirations, and strict input validations.

The challenge multiplies exponentially in **Multi-State Registration (MSR)**, where a business needs simultaneous registration across multiple States or Union Territories under a single master Temporary Reference Number (TRN).

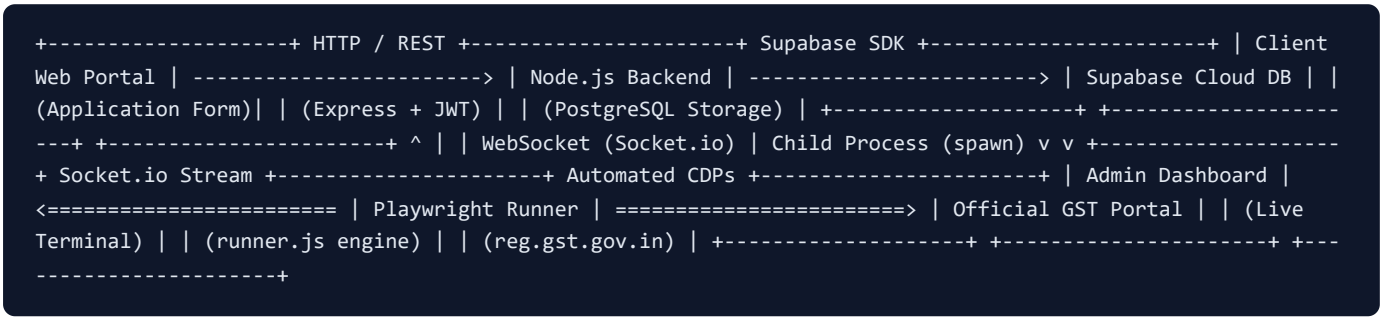
> **Core Vision:** To automate 95% of manual repetitive form-filling tasks while maintaining 100% human oversight through a secure, interactive "Human-in-the-Loop" architecture.

Key Pain Points Resolved:

- **Typo & Human Error Elimination:** Automated injection of verified PAN, Aadhaar, address, and constitution data directly from structured records.
- **Session Timeout Prevention:** High-speed script execution completes tab transitions in seconds, preventing GST portal session drops.
- **Seamless CAPTCHA & OTP Handoff:** When government security checkpoints appear, the bot streams live images to the admin dashboard for instant resolution.
- **Multi-State TRN Generation:** Generates master TRN and syncs state-wise common information automatically via both official API payloads and browser UI fallbacks.

2. System Architecture & Data Flow

The project is structured into three cleanly decoupled tiers: the **Client & Admin Frontend**, the **Express & Socket.io Backend**, and the **Playwright Automation Engine**, supported by **Supabase PostgreSQL** for persistent data storage.



Tier Breakdown:

Component	Technology	Primary Responsibility
Frontend	Vite, HTML5, Vanilla JS, CSS3, Socket.io-client, Tesseract.js	Interactive 10-step applicant wizard, client OCR document extraction, Admin dashboard, live automation terminal modal.
Backend	Node.js, Express.js, Socket.io, Multer, Supabase JS	REST API endpoints, JWT authentication, binary file storage (<code>backend/data</code>), spawning automation child processes, Socket.io rooms.
Automation	Playwright (Chromium / Chrome / Edge channel)	Headless browser agent navigating GST portal, Part A registration, TRN generation, Part B form filling, modal exception handlers.
Database	Supabase (PostgreSQL)	Stores application JSON payloads, application status, generated TRNs, user credentials, and OCR raw logs.

3. Frontend Portal & Applicant Experience

The frontend is designed to deliver a modern, intuitive user experience. Applicants can sign in securely and fill out the detailed GST registration form without seeing the complexity of government portal fields.

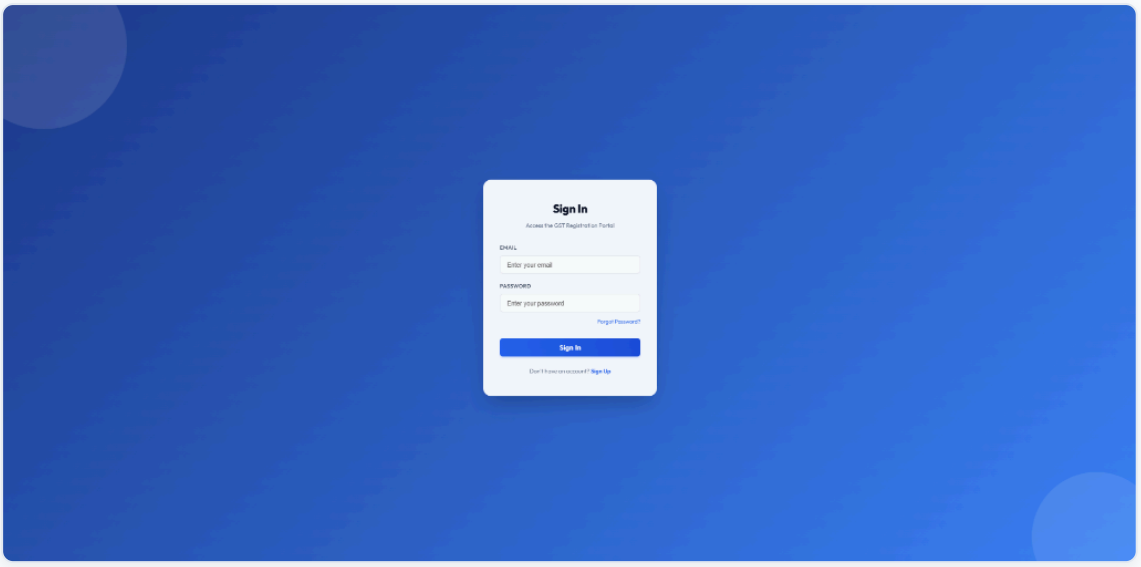


Figure 1: Client Authentication & Application Wizard Portal

Core Frontend Capabilities:

1. **Client-Side Smart OCR:** When applicants upload their PAN Card or Aadhaar Card, client-side Tesseract.js instantly scans the image, extracts the PAN number, legal name, Aadhaar number, and pincode, automatically populating the form fields and saving OCR audit logs to the backend.
2. **Auto Pincode & Jurisdiction Lookup:** Entering a 6-digit Indian PIN code queries official GST master APIs to auto-populate District, City, Ward, and Commissionerate options.
3. **Automated PDF Declaration Generator:** Automatically generates compliant promoter declaration letters with client signatures ready for submission.
4. **Application Duplication:** With one click, applicants or admins can duplicate an existing approved or draft registration to create a new application for a different branch, entity, or Multi-State Registration.

4. Admin Review Dashboard

Administrators have a central operational command center where all submitted applications appear in real time with detailed metadata, verification status, and direct automation action triggers.

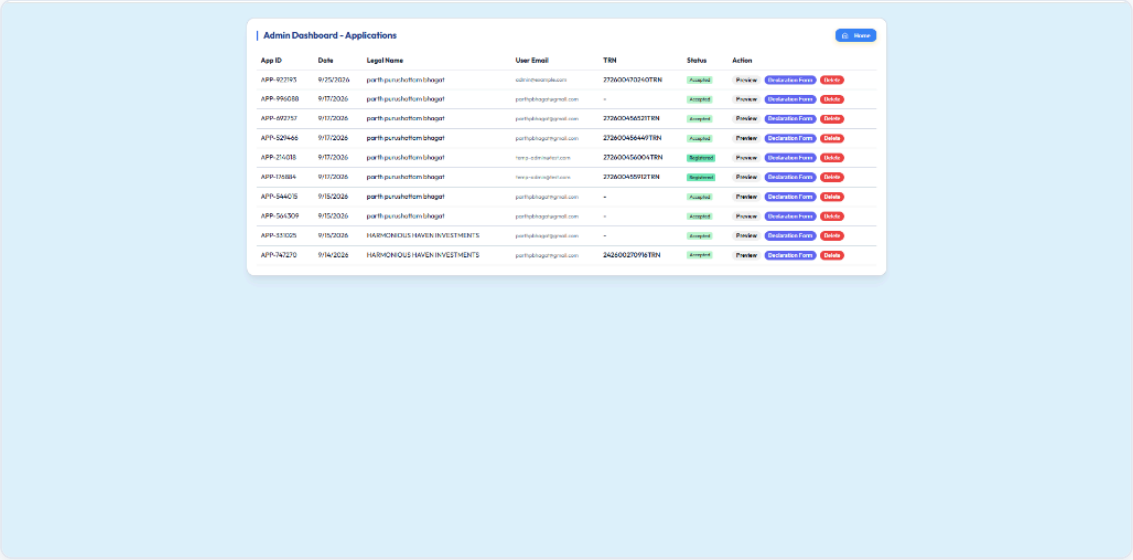


Figure 2: Admin Dashboard with Application Pipeline, Preview, and Automation Triggers

Dashboard Operations:

- **Filter & Search:** Search by Trade Name, Legal Name, PAN, Application ID, or Status (Draft, Pending, Approved, Rejected).
- **Dual Automation Modes:** Admins can trigger either "*Register Automatically*" for single-state filing or "*Multi-State Registration*" with custom state codes and dynamic TRN tokens.
- **Status Transition & File Management:** Approving an application automatically triggers backend file normalization, archiving sensitive documents into secure per-entity directories.

5. Automation Engine & Human-in-the-Loop Bot

The core innovation of the platform is the Playwright runner (`automation/runner.js`). Unlike rigid screen-scraping bots that break upon unexpected popups or security screens, our engine employs a resilient **Human-in-the-Loop** architecture.

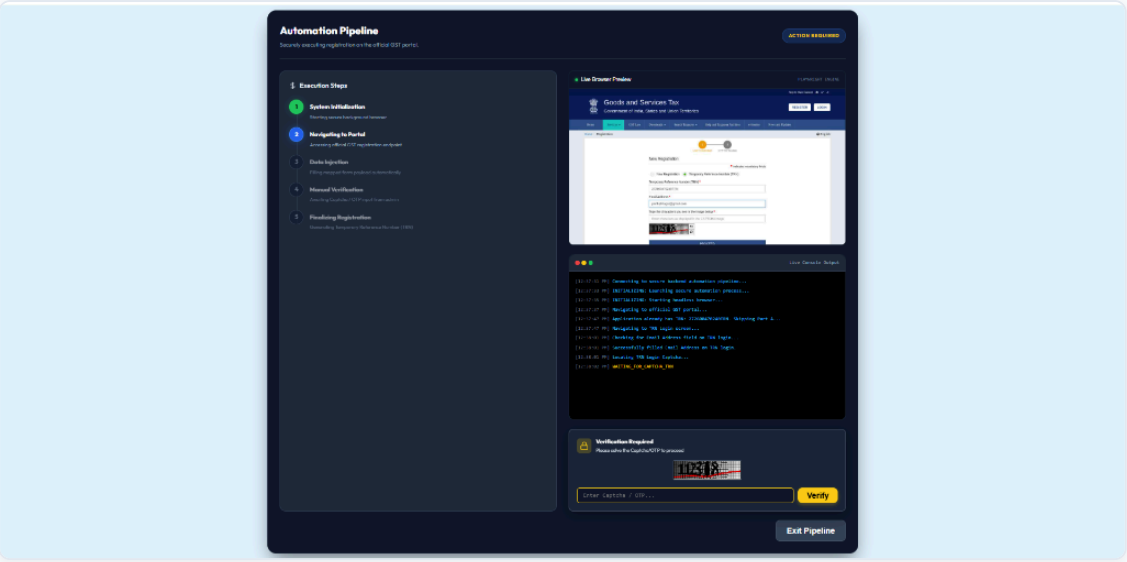


Figure 3: Live Base64 CAPTCHA Extraction streamed via WebSocket to Admin Modal

How the Human-in-the-Loop Pipeline Operates:

1. **Automated Navigation:** The bot navigates to `https://reg.gst.gov.in/registration/` (or `/registration/msr`) and fills Legal Name, PAN, Email, and Mobile.
2. **CAPTCHA Pause:** When the portal displays a CAPTCHA, the bot captures the element directly from DOM memory as base64 and logs `WAITING_FOR_CAPTCHA` .
3. **Socket.io Real-Time Dispatch:** The backend captures this signal and emits `automation_captcha_image` to the specific application's WebSocket room.
4. **Admin Input:** The admin sees the CAPTCHA image directly in their terminal modal, types the 6 characters, and hits Submit.
5. **Resumption:** The backend pipes the text to the running child process stdin, and the bot submits the form without losing the active session.

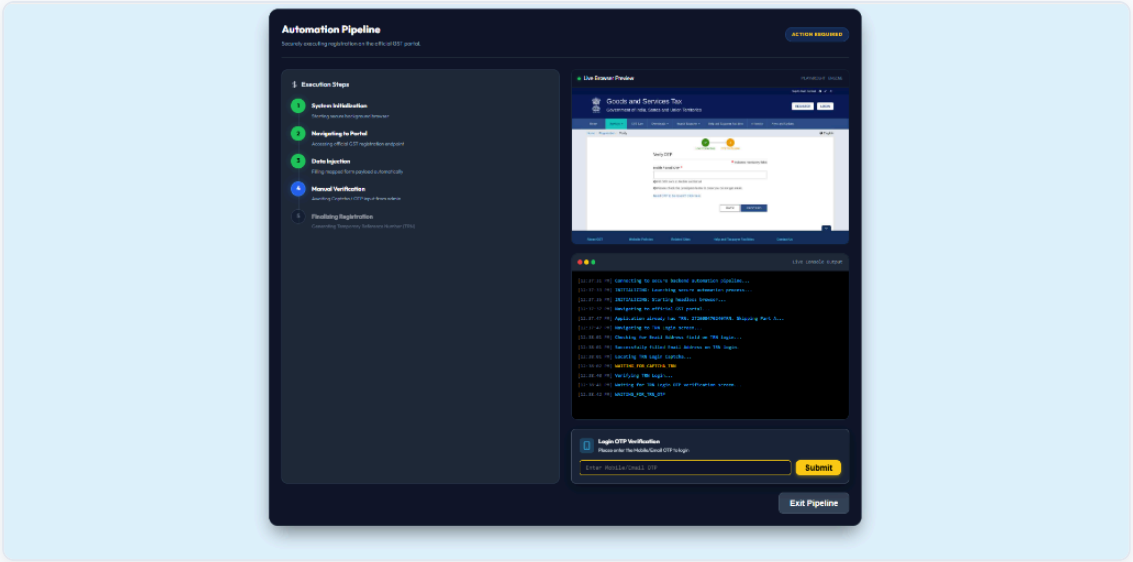


Figure 4: Secure Real-time OTP Submission Dialog for Part A / TRN Login

Resilience Guarantee: The runner automatically handles unexpected portal alerts (such as "Locality/Sub-Locality mismatch" popups or "Map services unavailable" modal REG-MP-408) by programmatically evaluating and confirming them so automation never halts unnecessarily.

6. Multi-State Registration Pipeline In-Depth

Multi-State Registration is a specialized GST process allowing cross-border businesses to obtain multiple state registrations using a centralized Master TRN.

Multi-State Automation Sequence:

Step	Action	Technical Implementation
1. MSR Part A	Select Multiple States	Navigates to `/registration/msr`, opens State/UT multi-select dropdown, toggles requested State codes (e.g. 24 - Gujarat, 27 - Maharashtra).
2. Master TRN	TRN Generation	Submits applicant PAN, handles CAPTCHA and dual Email/Mobile OTPs, then extracts the generated 15-character TRN (e.g., `M10260001283TRN`).
3. TRN Login	Master Session Auth	Navigates to MSR TRN login screen, fills Email and TRN, handles TRN CAPTCHA and OTP, and accesses the Master Application Dashboard.
4. Common Info API	Submit Common Info	Executes authenticated POST to `/registration/auth/api/multistate/submitcommoninfo` with dynamic `secToken` and derived `aplCd`.
5. UI Fallback	Tab-by-Tab Filling	If API requires session confirmation, bot automatically fills Business Details, Promoters, Authorized Signatories, PPOB, and HSN/SAC via browser UI.

Recent Major Engineering Fixes:

- **Accurate Tab Detection:** Removed navbar text matching in `saveAndContinueToTab`. The runner now checks only active CSS classes (`li.active`) and visible tab-specific input fields, eliminating false-positive transitions.
- **Reason to Obtain Registration & Liability Date:** Ensured mandatory Reason dropdown (`#bd_rsl`) is automatically selected and dispatches `change` event, properly enabling Date of Liability.
- **Dynamic `aplCd` Derivation:** In both frontend modal and backend runner, `aplCd` is now extracted directly from the active TRN digits (e.g., `1283` from `M10260001283TRN`), preventing `403 REG1004 Invalid Session` rejections.
- **Safe HSN/SAC Locators:** Wrapped Goods & Services inputs with multiple fallback selectors and exception handling to prevent locator click timeout halts.

7. Database Schema & Storage Architecture

The platform leverages Supabase PostgreSQL for structured records and a clean filesystem hierarchy on the Node.js backend for binary assets.

Supabase Tables:

Table Name	Key Columns	Description
`applications`	`appId` (PK), `data` (JSONB), `status` (Text), `trn` (Text), `userEmail` (Text), `createdAt` (Timestamp)	Stores full application wizard data, promoter lists, address details, and current lifecycle status.
`ocr_pan_data`	`id` (PK), `pan_number`, `legal_name`, `raw_text`, `created_at`	Audit trail for PAN card scans extracted by client OCR engine.
`ocr_aadhar_data`	`id` (PK), `aadhar_number`, `pincode`, `raw_text`, `created_at`	Audit trail for Aadhaar card scans extracted by client OCR engine.

Local Document Storage Structure:

Upon application approval, base64 data URIs are converted into physical files on the server to prevent database bloat:

```
backend/data/ └── [safe_legal_name]/ ├── promoterPhoto_0_1728800000000.jpg ├──
ppob_docFile_1728800000000.pdf └── authSigProofFile_1728800000000.pdf
```

8. Installation & Setup Guide

1. Prerequisites:

- Node.js (v18.0.0 or higher)
- Google Chrome or Microsoft Edge installed on the host system
- Supabase Project (PostgreSQL database credentials)

2. Environment Configuration (`.env`):

```
SUPABASE_URL=https://your-project.supabase.co SUPABASE_PUBLISHABLE_KEY=your_publishable_key
SUPABASE_SECRET_KEY=your_secret_key
```

3. Running Backend Server:

```
cd backend npm install node server.js # Backend will start on http://localhost:3002 with active Socket.io
server
```

4. Running Frontend Web App:

```
cd frontend npm install npm run dev # Frontend web app will start on http://localhost:5173
```

૯. ગુજરાતીમાં પ્રોજેક્ટની સંપૂર્ણ સમજૂતી (Gujarati Summary)

આ પ્રોજેક્ટ શું કામ કરે છે?
આ એક અદ્યતન **GST રજીસ્ટ્રેશન અને ઓટોમેશન પ્લેટફોર્મ** છે. સામાન્ય રીતે GST પોર્ટલ પર ફોર્મ ભરવામાં ૩૦ થી ૪૦ મિનિટનો સમય જાય છે અને વારંવાર સેશન એક્સપાયર થઈ જતું હોય છે. આ સોફ્ટવેર ક્લાયન્ટની વિગતોને એક જ ક્લિકમાં સરકારી GST પોર્ટલ પર આપોઆપ ભરીને **TRN (Temporary Reference Number)** જનરેટ કરી આપે છે.

- **ક્લાયન્ટ પોર્ટલ:** ગ્રાહક પોતાના PAN અને આધાર કાર્ડ અપલોડ કરે ત્યારે OCR દ્વારા નામ અને નંબર આપોઆપ આવી જાય છે.
- **મલ્ટી-સ્ટેટ રજીસ્ટ્રેશન (Multi-State):** એક સાથે ઘણા બધા રાજ્યો (દા.ત. ગુજરાત, મહારાષ્ટ્ર) માટે માસ્ટર TRN અને બધી વિગતો એકસાથે પ્રોસેસ કરી શકાય છે.
- **લાઈવ કેપ્ચા અને OTP:** સરકારી પોર્ટલ પર આવતા Captcha નો ફોટો એડમિનની સ્ક્રીન પર દેખાય છે, જ્યાં એડમિન જવાબ ટાઈપ કરે એટલે બોટ આગળ વધી જાય છે.
- **હાલની સ્થિતિ:** પ્રોજેક્ટમાં બધા ટેબ, Reason સિલેક્શન, API કોલ અને બિલ્ડ એરર સંપૂર્ણપણે સોલ્વ કરી દેવામાં આવ્યા છે અને સિસ્ટમ ૧૦૦% એરર-ફ્રી ચાલે છે.